

Improving Uncertainty Communication in Scientific Visualization

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Abstract

This document builds on the Research Proposal Draft (Milestone #2) and Annotated Bibliography (Milestone #4) to lay out a concrete plan for testing how different uncertainty encodings affect decision accuracy and confidence calibration. We identify a gap in head-to-head comparisons of error bars, distributional summaries, and hypothetical outcome plots (HOPs) within scientific visualization, and describe the experimental design, project timeline, and writing schedule needed to complete the research and paper in the remaining course weeks.

1 Introduction

Scientific visualization routinely presents results that come from measurement, modeling, or sampling—all of which carry uncertainty. When that uncertainty is left out or poorly conveyed, viewers can draw overconfident conclusions or misjudge how strong the evidence really is [1]. This project sits within the graphics and visualization research area, and the core question is practical: which uncertainty encodings actually help people interpret data more accurately and with better-calibrated confidence?

1.1 Purpose and Justification

Surveys have mapped where uncertainty enters the visualization pipeline [1], and general frameworks for encoding it in statistical graphics have been proposed [2], but head-to-head comparisons of common encodings—error bars, distributional summaries (box/violin plots), and hypothetical outcome plots (HOPs)—are still uncommon in scientific visualization settings. Stokes et al. [3] found that visualization outperforms text for rational decision-making, and work by Saklani et al. [4] and Li et al. [5] shows that folding uncertainty into the visualization makes scientific outputs

more trustworthy. Even so, there is a clear gap: very few studies pit error bars, distributional summaries, and HOPs against each other on the same tasks while measuring both accuracy and confidence calibration. We aim to fill that gap by running a controlled comparison, focusing not just on whether people get the right answer but on whether their confidence tracks their actual correctness—and whether richer encodings cost extra time.

1.2 Definition of Terms

- **Uncertainty:** Variability or imprecision in data arising from measurement error, sampling, or model estimation.
- **Error bars:** Graphical representation of uncertainty as intervals (e.g., 95% confidence intervals) around point estimates such as means.
- **Distributional summaries:** Visual encodings that depict the full or approximate distribution of data (e.g., box plots, violin plots).
- **Hypothetical outcome plots (HOPs):** Animated or static displays of multiple plausible outcomes drawn from the underlying distribution, used to convey variability.
- **Confidence calibration:** The degree to which a decision maker’s stated confidence aligns with actual correctness; well-calibrated individuals are more confident when correct and less confident when incorrect.
- **Decision accuracy:** The proportion of correct responses on tasks requiring interpretation of visualized uncertainty.
- **Scientific visualization:** The use of visual representations to explore and communicate scientific data, often involving spatial or statistical summaries.

2 Research Question

The primary research questions this study will address are:

1. Which uncertainty visualization encoding (error bars, distributional summaries, or HOPs) best improves users’ decision accuracy in scientific visualization tasks?
2. Which encoding best supports confidence calibration—i.e., alignment between stated confidence and actual correctness?

3. Do distributional encodings (box/violin, HOPs) incur a measurable cost in task completion time compared to error bars?

In short, we want to run a controlled, within-subjects comparison of three widely used encodings on standardized tasks, using the Palmer Penguins dataset as stimulus material, and see which one actually helps people the most.

3 Hypothesis Statement

Compared to traditional error bars, uncertainty encodings that show distributional variation (e.g., hypothetical outcome plots or distribution summaries such as box/violin plots) will produce higher decision accuracy and better confidence calibration without increasing task time.

3.1 Anticipated Outcomes

If the hypothesis holds, we should see higher mean decision accuracy when participants use distributional or HOPs displays compared to plain error bars. We also expect better confidence calibration—measured by Brier score or calibration curves—in those same conditions, meaning people’s stated confidence should line up more closely with whether they actually got the answer right. Finally, we predict that this improvement will not come at the cost of significantly longer task completion times. If all three outcomes hold, it would suggest that giving viewers richer distributional information helps them interpret data without adding prohibitive cognitive load.

3.2 Scope and Limitations

Scope: The investigation is limited to static and animated 2-D statistical graphics (bar charts with error bars, box plots, violin plots, and HOPs-style displays) using a single publicly available dataset (Palmer Penguins). Tasks will focus on ranking, estimation, and decision selection. Participants will be recruited from a convenience sample (e.g., students, colleagues) rather than domain scientists.

Limitations: Results may not generalize to other datasets, domains (e.g., volume visualization [4], trajectory visualization [5]), or populations. The study does not address multimodal uncertainty communication (speech, text) as in Stokes et al. [3]. Sample size will be constrained by the course timeline; power analysis will inform interpretability of null results.

4 Review of Literature (Updated Annotated Bibliography)

Uncertainty visualization has been an active area of research for some time now. This section reviews the key sources from our annotated bibliography and pulls together how each one informs the study we are planning.

4.1 Introduction to the Literature

Uncertainty can creep in at every stage of the visualization pipeline—data collection, preprocessing, the visualization itself, and the inferences viewers draw from it [1]. The challenge is twofold: authors need practical ways to encode uncertainty, and viewers need to be able to interpret those encodings correctly. Recent papers have tackled encoding design [2], run empirical comparisons across communication modes [3], and applied uncertainty-aware methods to scientific domains like volume rendering and particle tracking [4], [5]. What is still missing is a systematic, apples-to-apples comparison of error bars, distributional summaries, and HOPs that measures both accuracy and calibration.

4.2 Summary of Sources

Mehta (2022) provides a broad survey of uncertainty in visualization [1]. Mehta breaks the visualization pipeline into four stages—data collection, preprocessing, visualization, and inference—and walks through how uncertainty affects each one. The survey also looks at how visualization authors currently handle uncertainty in practice, from choosing encodings to dealing with missing data. We *borrow* Mehta’s pipeline framework to scope our study: we hold the data-collection and preprocessing stages constant (using the same Palmer Penguins dataset across conditions) and isolate the visualization stage, where encoding choice matters most. We also *extend* the survey’s call for more empirical work by designing a controlled experiment that directly measures how encoding affects viewer accuracy and calibration.

Petek et al. (2025) take a more formal angle [2]. They treat a visualization as a function of some underlying quantities, and when those quantities are uncertain the function produces a distribution over images rather than a single image. From that framing, familiar representations like confidence intervals fall out naturally with coverage guarantees. They also release an open-source Python implementation. We plan to *borrow* their resampling-based image-generation approach to build the distributional and HOPs-style stimuli for our study, since their library can produce overlay plots with minimal custom code. We will also *test* a core implication of their framework: that richer distributional displays should carry more information than simple error bars, and we will measure whether that extra information actually translates into better viewer performance.

Stokes et al. (2024) shift the focus to how different communication modes—speech, text, and visualization—convey data uncertainty [3]. Across two crowdsourced experiments they found that visualization and text were the most effective for rational decision-making, but text lowered confidence while speech inspired the most trust (sometimes at the cost of riskier choices). We *borrow* two elements from their design: the paired measurement of decision accuracy alongside confidence, and the use of crowdsourced tasks for scalable data collection. We *extend* their work by holding the modality constant (all visual) and varying the encoding within that modality—something Stokes et al. flag as an open direction but do not pursue themselves.

The last two sources ground the project in scientific visualization practice. **Saklani et al. (2024)** build uncertainty-aware implicit neural representations for volume rendering and show, using Deep Ensembles and Monte Carlo Dropout, that surfacing prediction uncertainty makes DNN-based visualizations more trustworthy for domain scientists [4]. Their finding that uncertainty-aware displays increase trust is something we want to *test* in a simpler setting: if showing uncertainty improves trust in 3-D volume rendering, does the same principle hold when comparing 2-D statistical graphics like error bars versus distributional plots? **Li et al. (2025)** tackle a different data type—particle trajectories—with an “uncertainty tube” that captures nonsymmetric uncertainty through a superelliptical cross-section [5]. We *challenge* the implicit assumption in domain-specific work like this that every data type requires a bespoke encoding; our study tests whether general-purpose encodings (error bars, box/violin, HOPs) already perform well enough on common tasks to serve as practical defaults. Together, these two papers show that the right representation depends on context, which further motivates a controlled comparison on a shared benchmark.

4.3 Synthesis and Conclusion

Taken together, the literature makes a few things clear. Uncertainty is everywhere in scientific data and ignoring it misleads viewers [1]. We now have formal tools for encoding it [2], and empirical evidence that the choice of representation changes how people decide and how confident they feel [3]. At the same time, applied work in volume rendering [4] and trajectory visualization [5] shows that the “best” encoding depends heavily on context. What is lacking is a direct, controlled comparison of the three most common encodings—error bars, distributional summaries, and HOPs—on the same tasks, measuring both accuracy and calibration. That is the gap our study is designed to address.

5 Research Plan

Below we expand the preliminary design from Milestone #2 into a concrete experimental plan, a project management schedule, and a writing plan. Everything is scoped to fit the remaining 8–10 weeks of the course, targeting a full draft by May 1 and a polished version by May 8.

5.1 Experimental Plan

5.1.1 Framework, Environment, and Tools

- **Visualization:** Python (Matplotlib, Seaborn) or R (ggplot2) for generating error bars, box plots, violin plots, and HOPs-style displays. HOPs will be implemented as static grids of bootstrap sample means (as in the preliminary design) or as animated sequences if time permits.
- **Study platform:** Qualtrics, Google Forms, or a lightweight web app (e.g., Streamlit) for presenting stimuli and collecting responses.
- **Analysis:** R or Python (pandas, scipy, statsmodels) for repeated-measures ANOVA, non-parametric tests (Friedman, Wilcoxon), and calibration metrics (Brier score, calibration curves).
- **Version control:** Git for code and data; Google Drive for the paper.

5.1.2 Data Usage

- **Primary dataset:** Palmer Penguins (bill length, flipper length, body mass by species). Data will be acquired from the public repository (e.g., seaborn-data on GitHub) and preprocessed to compute bootstrap confidence intervals and sample distributions.
- **Acquisition:** Download once; store locally with a clear README documenting source and preprocessing steps.
- **Management:** CSV files for summary statistics; reproducible scripts for generating all figures and derived data. No personally identifiable information.

5.1.3 Test Methodologies and Evaluation Metrics

- **Methodology:** Within-subjects design. Each participant sees all three encodings (error bars, box/violin, HOPs) in counterbalanced order. Tasks: (1) rank groups by a statistic; (2) estimate a difference; (3) select the most reliable trend. After each task, participants report confidence (e.g., 0–100%).

- **Metrics:** (1) Decision accuracy (proportion correct); (2) Confidence calibration (Brier score, calibration curves); (3) Task completion time (seconds); (4) Optional: perceived clarity and trust (Likert items).
- **Analysis:** Repeated-measures ANOVA or Friedman test for accuracy and time; paired comparisons for calibration. Effect sizes (e.g., partial η^2) will be reported.

5.1.4 Test Plan

- **What:** Three encoding conditions $\times$ three task types $\times$ multiple trials per condition. Pilot with 3–5 participants to refine prompts and catch technical issues.
- **How:** Online survey with embedded images; participants complete tasks at their own pace. Attention checks and consent form included.
- **When:** Pilot in Week 2 of implementation; full data collection in Weeks 3–4; analysis in Weeks 5–6.

5.2 Project Management and Feasibility

5.2.1 Week-by-Week Timeline

- **Week 1:** Finalize stimuli (all three encodings for Palmer Penguins), build survey instrument, obtain IRB/ethics approval if required.
- **Week 2:** Run pilot; revise prompts and interface; begin recruitment.
- **Week 3–4:** Data collection; target $N \geq 24$ for within-subjects power (adjust based on pilot).
- **Week 5:** Data cleaning, descriptive statistics, inferential analysis.
- **Week 6:** Complete analysis; draft Results and Discussion; create figures for paper.
- **Week 7:** Integrate all sections; first full draft by May 1.
- **Week 8:** Revise, polish, proofread; polished version by May 8.
- **Weeks 9–10 (if applicable):** Presentation preparation; final revisions.

5.2.2 Anticipated Challenges and Risks

- **Low recruitment:** Convenience sampling may yield insufficient participants.
- **Technical failures:** Survey platform or display issues could invalidate data.

- **Null results:** Small effects or high variance may preclude clear conclusions.
- **Time overrun:** Analysis or writing may take longer than planned.

5.2.3 Risk Mitigation and Fallback Options

- **Recruitment:** Broaden recruitment (classmates, departmental lists, social media); reduce target N and frame as exploratory if necessary.
- **Technical:** Test survey on multiple devices; include fallback instructions (e.g., “If the image does not load, please refresh”).
- **Null results:** Report effect sizes and confidence intervals; discuss limitations and future work; emphasize methodological contribution.
- **Time:** Prioritize core analysis and writing; defer optional metrics (e.g., perceived trust) if needed. Draft each section as work progresses rather than deferring all writing to the end.

5.3 Writing Plan

5.3.1 Outline of the Research Paper

1. **Introduction:** Motivation, gap, contribution, roadmap.
2. **Background / State of the Art:** Uncertainty visualization; prior comparisons of encodings; calibration.
3. **Problem Formulation and Motivation:** Research questions, hypothesis.
4. **Solution / Method:** Dataset, encodings, tasks, procedure, participants, analysis plan.
5. **Related Work:** Condensed literature review (can draw from Section 4 of this plan).
6. **Proof / Evaluation / Discussion:** Results, figures, statistical tests, interpretation, limitations.
7. **Conclusion:** Summary, implications, future work.
8. **Future Work:** Extensions (other datasets, domains, populations).

5.3.2 Schedule for Drafting, Revising, and Polishing

- **Weeks 1–2:** Draft Introduction and Background (draw from this plan).
- **Week 3:** Draft Method section as design is finalized.

- **Weeks 5–6:** Draft Results and Discussion as analysis completes.
- **Week 7:** Integrate all sections; complete first full draft by **May 1**.
- **Week 8:** Revise for clarity and flow; polish prose; proofread; final version by **May 8**.

5.3.3 Coverage of Expected Components

The outline above covers every component expected in the final paper: Introduction, Background, Problem Formulation, Solution/Method, Related Work, Evaluation/Discussion, Conclusion, and Future Work. The plan is to draft each section as the corresponding work is done rather than saving all the writing for the end. That way, Week 8 is purely for revision and polishing instead of a last-minute scramble.

6 References

- [1] K. Mehta, “Maybe, Maybe Not: A Survey on Uncertainty in Visualization,” arXiv preprint arXiv:2301.07687, 2022. [Online]. Available: <https://arxiv.org/abs/2301.07687>
- [2] B. Petek, D. Nabergoj, and E. Štrumbelj, “A General Approach to Visualizing Uncertainty in Statistical Graphics,” arXiv preprint arXiv:2508.00937, 2025. [Online]. Available: <https://arxiv.org/abs/2508.00937>
- [3] C. Stokes, C. Sanker, B. Cogley, and V. Setlur, “From Delays to Densities: Exploring Data Uncertainty through Speech, Text, and Visualization,” *Computer Graphics Forum*, vol. 43, no. 3, 2024. (Presented at EuroVis 2024; published by Wiley.) Available: <https://arxiv.org/abs/2404.02317>
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- [5] J. Li, T. A. J. Ouermi, M. Han, and C. R. Johnson, “Uncertainty Tube Visualization of Particle Trajectories,” in *Proc. IEEE Workshop Uncertainty Vis. (IEEE VIS)*, Vienna, Austria, Nov. 2025. Available: <https://arxiv.org/abs/2508.13505>